



RIVER VALLEY HIGH SCHOOL

JC 2 PRELIMINARY EXAMINATION

CANDIDATE
NAME

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CENTRE
NUMBER

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CLASS

21J

INDEX
NUMBER

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BIOLOGY

9744/02

Paper 2 Structured Questions

12 September 2022

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE ON ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	15
2	8
3	9
4	10
5	10
6	10
7	8
8	11
9	9
10	5
11	5
Total	100

Answer **all** questions.

- 1 In the lung, epithelial cells have a thin layer of watery mucus on their surface to trap particles.

Fig. 1.1 shows a series of processes that occur at the cell surface membrane of an epithelial cell, involving a channel protein called the Cystic Fibrosis Transmembrane conductance Regulator (CFTR).

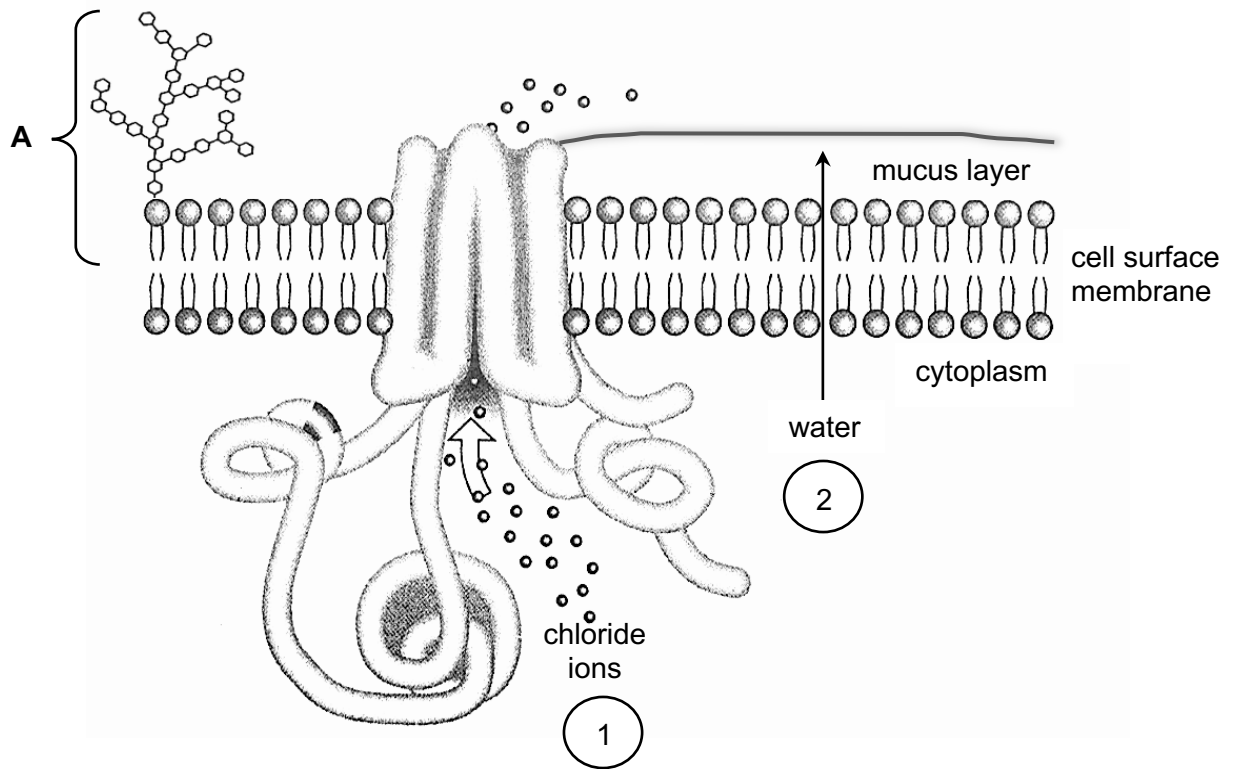


Fig. 1.1

- (a) Identify the structure labelled **A** and state its role. [2]

structure **glycolipid;;**

AOA

role **Cell surface marker for cell-cell recognition / cell-cell adhesion to form tissues;;**

- (b) (i) Explain why chloride ions can only cross cell surface membranes via CFTR. [3]

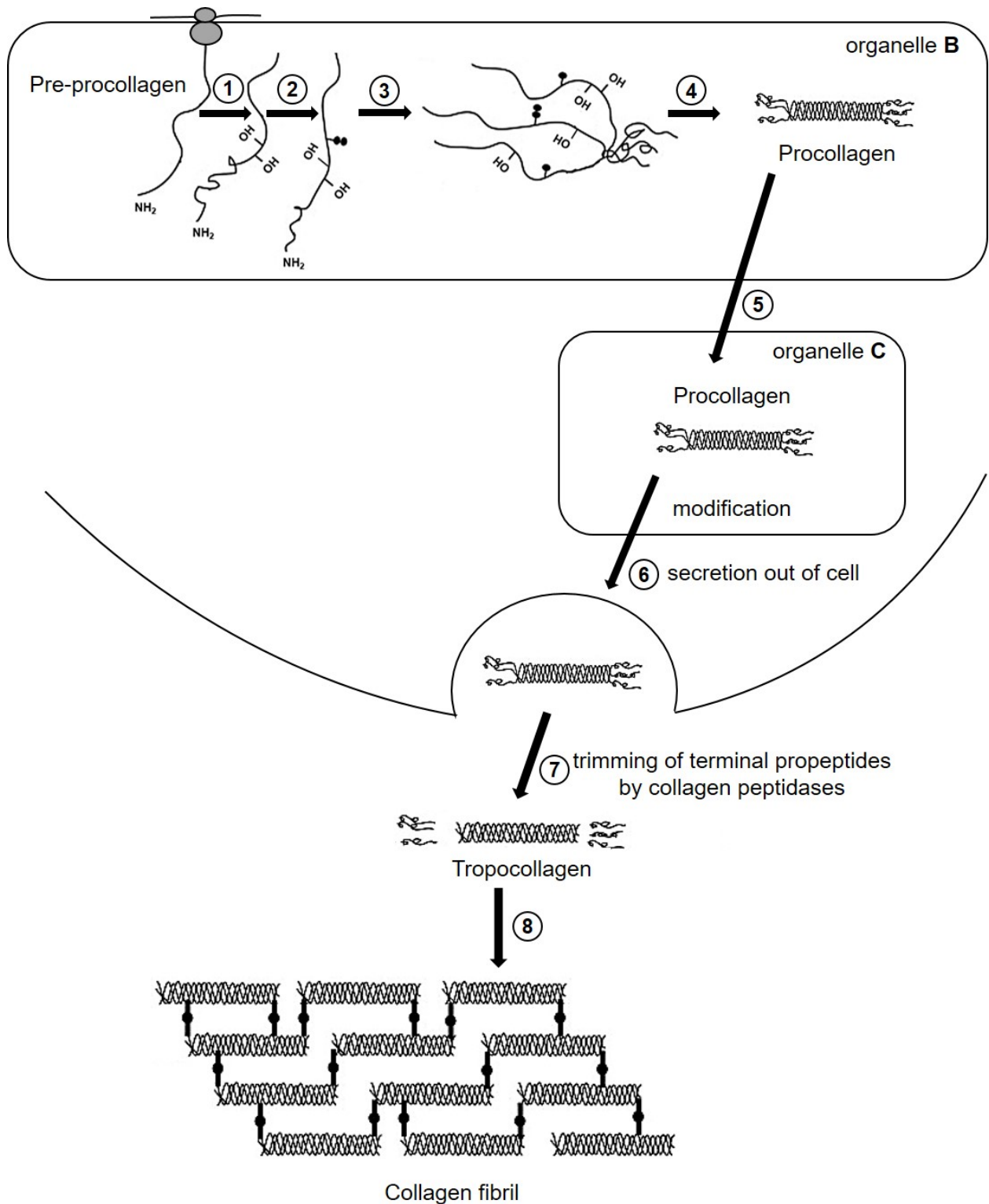
1. **chloride ions are charged;;**
2. **cannot diffuse through hydrophobic core of cell surface membrane;**
3. **CFTR provides a hydrophilic channel that shields Cl^- from the hydrophobic core;**

4. the channel is specific to the transport of chloride ions as it has a specific binding site complementary to the shape of chloride ions;;

(ii) Explain why absence of CFTR results in more viscous mucus. [2]

1. chloride ions remain inside epithelial cell;
2. water potential inside the cell is more negative than outside the cell;;
3. water enters the cell via osmosis;
resulting in more viscous mucus

Fig. 1.2 shows the stages during the synthesis of collagen.



key

OH: hydroxyl group

• : sugar

Fig. 1.2

- (c) (i) Identify organelles **B** and **C** in Fig. 1.2. [2]

organelle **B** **Rough endoplasmic reticulum;;**

organelle **C** **Golgi apparatus;;**

- (ii) With reference to Fig. 1.2, describe how tropocollagen is synthesised from pre-procollagen. [3]

1. **Addition of OH groups / hydroxylation;**
2. **Addition of sugars / glycosylation;**
3. **Three pre-procollagen chains joined together to form procollagen in organelle B (RER);**
4. **Procollagen transported to organelle C (GA) to be modified;**
5. **secreted out of the cell;**
A: released via exocytosis
6. **Terminal propeptides are cleaved / removed / trimmed by collagen peptidases;**

- (iii) The feet of elephants are covered in padding to support the mass of the elephant. This padding is made up of a large number of cells surrounded by connective tissue containing many fibres of collagen.

Explain how **stage 8** contributes to the function of the padding. [2]

1. **Tropocollagen molecules lie parallel to each other in a staggered arrangement;**
2. **joined by covalent bonds between carboxyl and amino ends / cross-linked;**
3. **bundling of fibrils to form fibres;**
4. **contribute to high tensile strength (to support the mass of the elephant);**

- (iv) Suggest why collagen fibril is assembled outside the cell. [1]

1. **Enzymes involved in the removal of propeptides are found outside the cell;;**
2. **Tropocollagen is too large to pass through the cell surface membrane;;**

Any one

[Total: 15]

- 2 Enzymes can be immobilised in alginate beads to improve their physical and biochemical properties. A study was carried out to investigate the effect of pH on the activity of free GH and immobilised GH. Equal concentrations of free GH and immobilised GH were used.

Fig. 2.1 shows the results of this study.

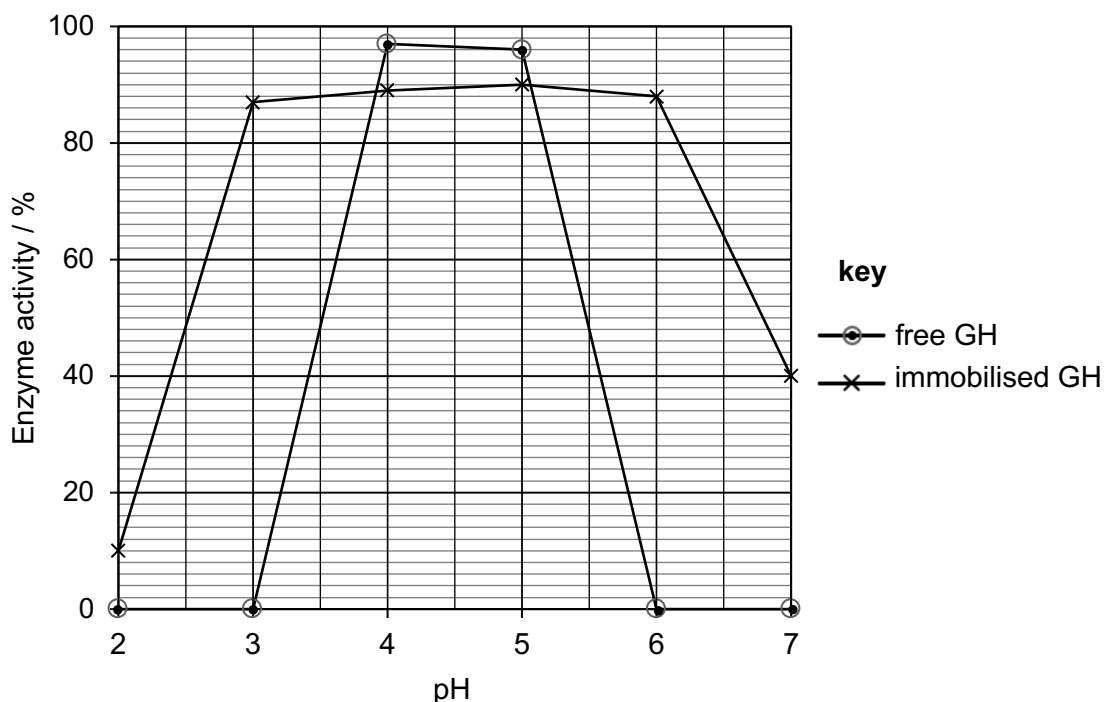


Fig. 2.1

- (a) With reference to Fig. 2.1, describe **two** differences between the activity of free GH and immobilised GH.

Suggest an explanation for each of the two differences.

[4]

difference 1 and explanation:

difference 2 and explanation:

Description

1. immobilised GH has a wider range of optimum pH than free GH;
 2. QV: optimum pH of immobilised GH is 3-6 but that of free GH is 4-5;
- or
3. immobilised GH is functional over a wider range of pH than free GH;
 4. QV: immobilised GH is 2 to 7 but that of free GH is >3 to <6;

Explanation

5. Immobilised GH less exposed to H^+ ;

6. Less change in charge of acidic and basic R group of amino acid residues / fewer breakage of hydrogen bonds and ionic bonds;

or

7. immobilisation restricts change in shape / unfolding of polypeptide chain;

8. despite breakage of ionic and hydrogen bonds;

Description

9. free GH has higher maximum activity than immobilised GH;

10. QV: max activity of free GH is 96-97% but that of immobilised GH is 87-90%;

Explanation

11. Increase accessibility;

12. Higher frequency of effective collision between active site of free GH and substrate / higher number of enzyme-substrate complex formed per unit time;

Glucoside hydrolase (GH) is an enzyme responsible for hydrolysing α -1,4-glycosidic bonds in carbohydrates, such as amylose and amylopectin.

(a) Using GH as an example, explain induced-fit hypothesis. [2]

1. Active site of glucosidase has a shape complementary to that of α -1,4 glycosidic bond;
2. initial shape of enzyme active site may not be completely complementary to the shape of substrate;
3. Binding of the substrate to the active site induces a conformational change in the enzyme;
4. This enables the substrate to fit more snugly into the active site (to form enzyme-substrate complex);

Fig. 2.2 shows a molecule of amylose and a molecule of RNA (not drawn to scale).

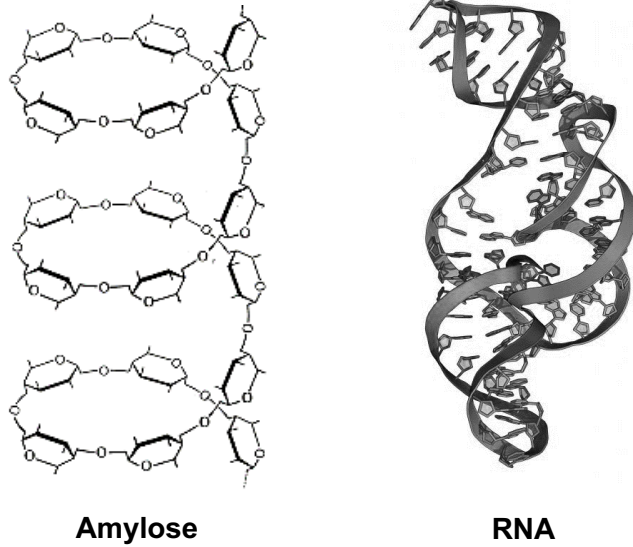


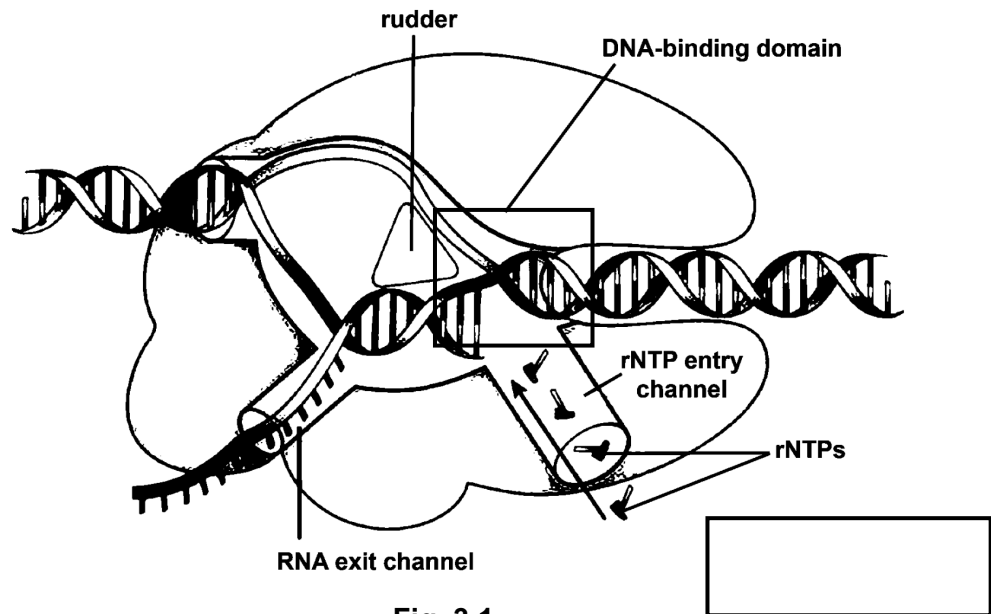
Fig. 2.2

(c) Describe in what ways the helical structures in both amylose and RNA differ. [2]

Feature	Amylose	RNA
1: Helical structure;;	Single helix	Double helix
2: Formation of helical structure;;	Folding / coiling within <u>one polysaccharide chain</u>	<u>Two segments of RNA molecule</u> folds back on itself
3: Bond forming helical structure;;	α -1,4-glycosidic bonds	H bonds <u>between RNA nucleotides</u> (complementary base pairing)
4: Regularity of helical structure;;	Regular (coils throughout molecule)	Irregular (helix formed at some parts of the molecule)

[Total: 8]

- 3 Fig. 3.1 shows a diagram of RNA polymerase during transcription of the β -globin gene.



- (a) (i) In the box on Fig. 3.1, label with an arrow the direction of transcription. [1]



- (ii) With reference to Fig. 3.1, describe how the structure of RNA polymerase allows it to perform its function. [2]

1. (S) DNA binding domain has a specific 3-dimensional conformation (P) complementary to the shape of DNA;

(F) allows RNA polymerase to bind to DNA / bind to promoter to initiate transcription;

2. (S) Presence of rudder;

(F) to separate the two DNA strands so as to expose template strand;

3. (S) Presence of rNTP channel (P) that is complementary to shape of rNTP;

(F) allows incoming ribonucleoside triphosphate (rNTP) to enter RNA polymerase for elongation;

4. (S) Presence of RNA exit channel;

(F) allows RNA strand to leave the RNA polymerase as it elongates;

Any two

Actinomycin D and cycloheximide are drugs used in the treatment of chronic leukemia, and are involved in the inhibition of β -globin synthesis.

Fig. 3.2 shows the results obtained when each drug is added to immature red blood cells in separate experiments. The thickness and intensity of the bands are an indication of the amount of β -globin mRNA or β -globin protein present. Results have been adjusted to allow for direct comparison.

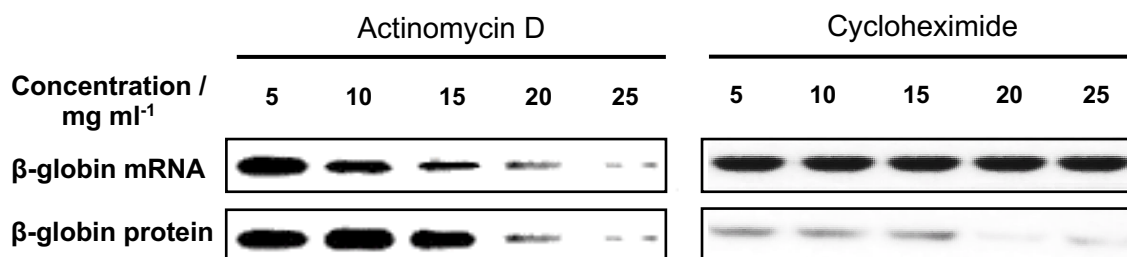


Fig. 3.2

(b) (i) State the drug that inhibits the following processes in β -globin synthesis. [1]

transcription **Actinomycin D**

translation **Cycloheximide**

(ii) Justify your answer to (b)(i). [3]

1. (Evidence) As concentration of actinomycin D added increases from 5 mg ml^{-1} to 25 mg ml^{-1} , both the amount of β -globin mRNA and β -globin protein decreases;;

2. therefore showing that less mRNA templates were produced for translation;

3. (Evidence) As concentration of cycloheximide added increases from 5 mg ml^{-1} to 25 mg ml^{-1} , amount of β -globin mRNA remains the same; but amount of β -globin protein decreases;

4. therefore showing that the available mRNA templates were not used for translation;

(c) Describe **two** features of the genetic code. [2]

1. consists of a triplet of bases (codons);;

2. universal;;

3. non-overlapping / mRNA sequence can be read continuously in a series of triplets without skipping any nucleotides;;

4. degenerate / more than 1 codons may code for the same amino acid;;

Any two

[Total: 9]

4 Cancer is a genetic disease. Despite this, only around 10% of all cancers are inherited.

(a) Suggest **two** reasons why a cancer that occurs in a parent does not always occur in the offspring. [2]

1. Mutations that result in the cancer occurs in somatic cells and not in germline cells, therefore absent in gametes;;
2. Mutations that result in the cancer may only occur after offspring has been born;;
3. Inheriting mutated gene only increases risk of developing the cancer;;
4. The cancer type is due to recessive alleles, may skip generation;;
5. AVP;;

Max 2

Acute myeloid leukemia (AML) is a rare and extremely rapid developing cancer in human. AML results from aberrant proliferation and differentiation of mutated myeloid cells in the bone marrow. Large numbers of malfunctioning differentiated cells then result in symptoms such as fatigue, unusual bruising, and a high number of infections.

Patients with AML display mutations in many genes but mutations in *FLT3* and *DNMT3A* genes are most prevalent. *FLT3* and *DNMT3A* are a type of receptor tyrosine kinase and DNA methyltransferase respectively.

Fig. 4.1 shows the development of AML in the bone marrow.

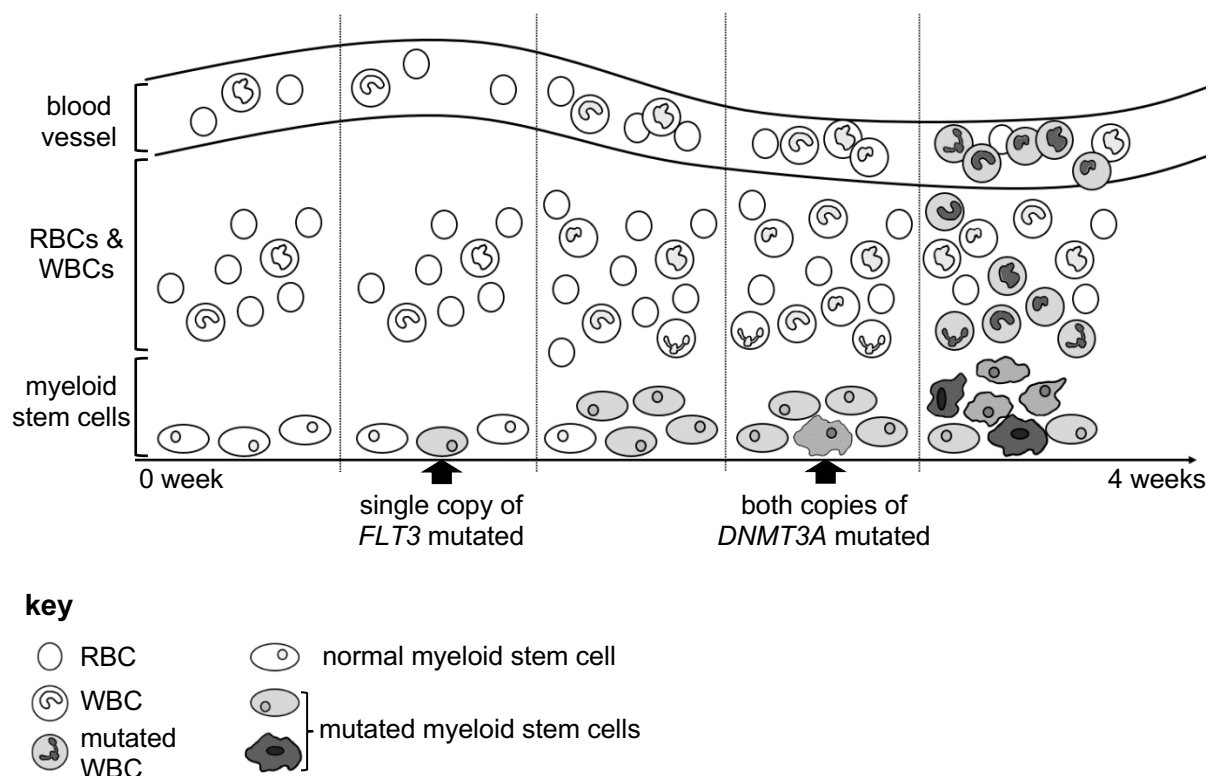


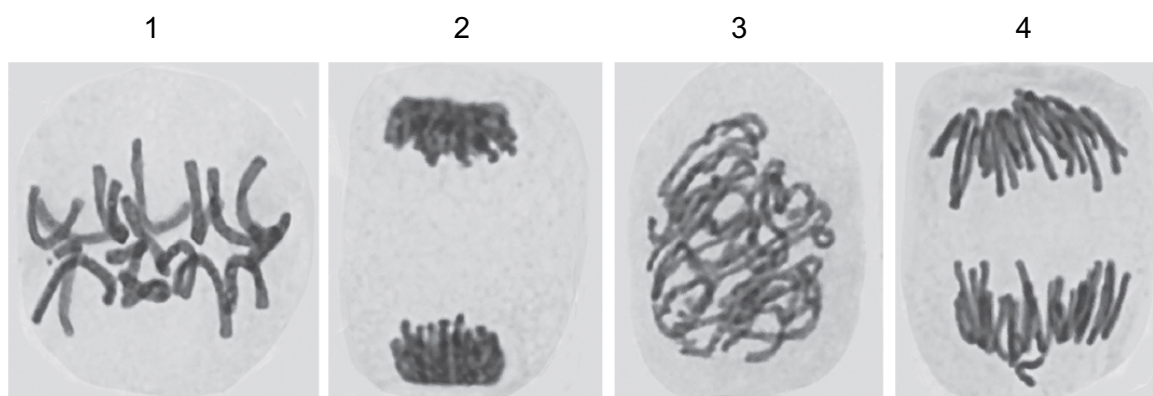
Fig. 4.1

- (b) Outline how the mutations in myeloid stem cells result in the changes visible in Fig. 4.1 and the formation of multiple secondary sites of tumour development. [4]

1. *FLT3* (proto-oncogene) undergoes gain-of-function mutation;
2. RTK becomes hyperactive / stimulates excessive cell division (to form more mutated myeloid cells);
3. *DNMT3A* (tumor suppressor gene) undergoes loss-of-function mutation;
4. Unable to inactivate gene expression of multiple genes / over-expression of many genes;
5. Results in uncontrolled cell division;
6. Differentiation (of mutated myeloid stem cells) to form more; malfunctioning WBCs than RBCs;
7. Metastasised via blood vessel to other parts of body;
8. mutated WBCs invade tissue and proliferate excessively;

Max 4

Myeloid stem cells were obtained from a patient at 0 week. The photomicrographs below show the myeloid stem cells in various stages of nuclear division.



- (c) (i) State which cell(s) contain twice as many chromosomes as a cell in resting phase from the same patient. [1]

2 and 4;;

- (ii) With reference to cells 1, 2 and 4, describe in correct order the behaviour of chromosomes in these stages of nuclear division. [3]

1. (in cell 1) chromosomes line up in a single row;
2. At the metaphase plate;
3. (in cell 4) separation of centromeres;
4. Sister chromatids separate to form daughter chromosomes;
5. Move to opposite poles of the cell;

6. (in cell 2) decondensation/uncoiling of chromosome to chromatin;

[Total: 10]

- 5 Haemagglutinin (HA) protein of influenza virus is a homotrimeric protein complex. Each subunit is coded for by the *HA* gene present on one of its eight genomic segments.

Fig. 5.1 shows one of the subunits.

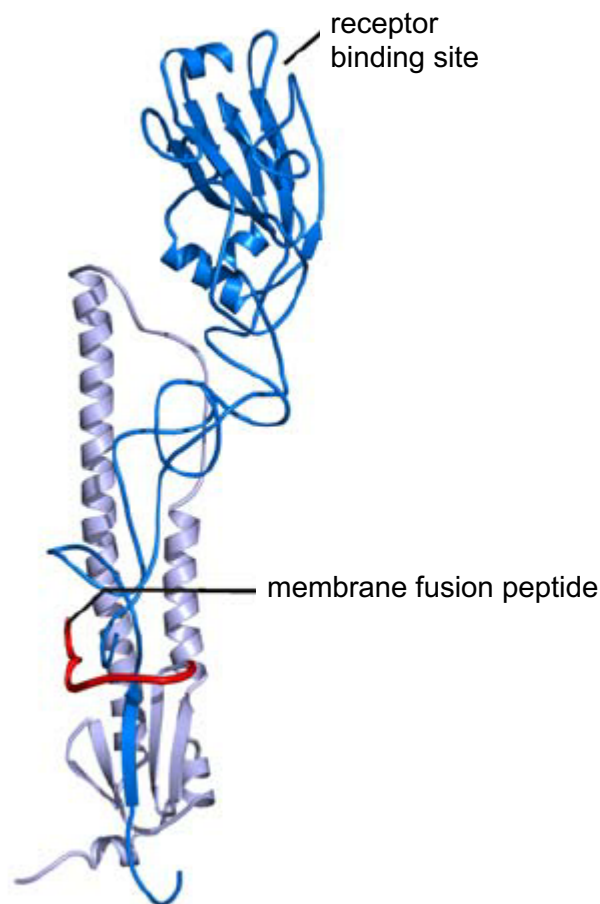


Fig. 5.1

- (a) Outline how the identified features in Fig. 5.1 facilitate influenza infection. [3]

1. Receptor binding site of HA binds to specific glycoprotein receptors/sialic-acid receptors;
2. on cell surface membrane of host cell;
3. Membrane fusion peptide helps viral envelope to fuse;
4. With membrane of endosome (when pH decreases);
5. Results in release of ssRNA (-) genome (and viral RNA-dependent RNA polymerase);
6. into host cell cytoplasm;

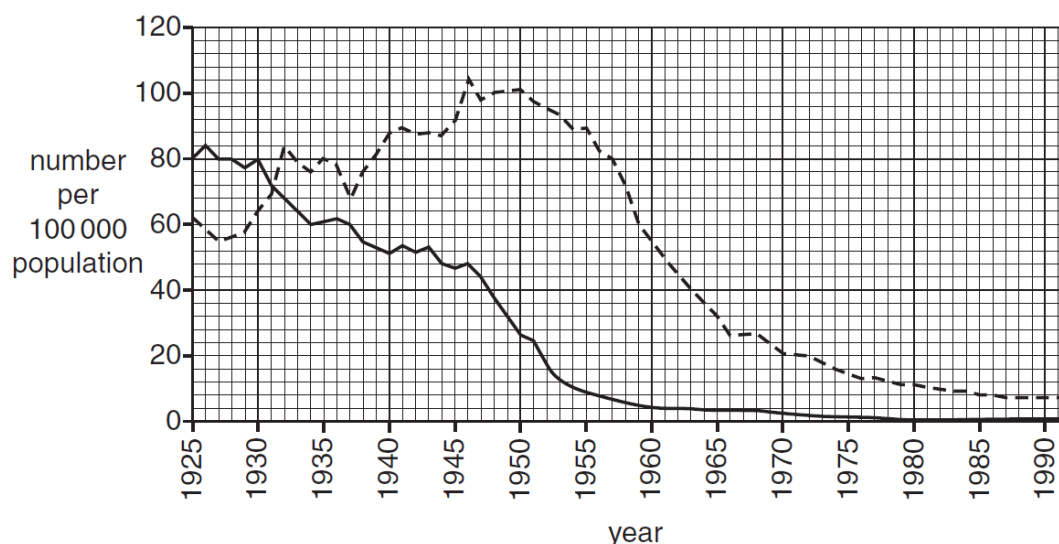
- (b) Explain how influenza virus with eight genomic segments can result in a pandemic. [2]

1. **Antigenic shift;**
2. **When two different strains infect same host cell;**
3. **Genetic recombination of RNA segments / rearrangement of genes;**
4. **During viral assembly;**
5. **Emergence of novel strain;**

Max 2

Fig. 5.2 shows the number of deaths from influenza and the number of new cases of influenza from 1925 to 1991 in Country K.

An antiviral drug for influenza was introduced in Country K for widespread use in 1930. The antiviral drug specifically targets non-envelope viral proteins to reduce the influenza's ability to multiply.



key

- deaths from influenza
 ----- new cases of influenza

Fig. 5.2

- (c) Using the information provided, account for the difference in the ten-year effect of the introduction of the antiviral drug on the number of deaths and number of new cases of influenza. [5]

The introduction of antiviral for influenza, from 1930 to 1940,

1. **Results in a decrease in number of deaths from 80 to 52 per 100 000 population;;**
2. **but an increase in number of new cases from 64 to 88 per 100 000 population;;**

Reduces number of deaths by

3. targeting RNA-dependent RNA polymerase;
 4. prevents use of ssRNA (-) as template;
 5. To synthesise complementary ssRNA (+);
 6. Reduces/no mRNAs for translation to form viral proteins;
 7. No templates to make new copies of ssRNA (-) genome;
 8. to prevent death of epithelial cells/apoptosis in respiratory tract/lungs;
9. But does not prevent influenza spread / infection;
 10. Through respiratory droplets between individuals;
- Max 5

[Total: 10]

- 6 Gene expression in eukaryotic cells is regulated at multiple levels.

Fig. 6.1 shows the regulation of transferrin receptor (TfR) expression using Iron Response Element-Binding Proteins (IRE-BP). IRE-BP binds to IREs which are loop structures found on the mRNA.

TfRs are present on the cell surface and are involved in the uptake of extracellular iron.

TfR mRNA in **cytosol**

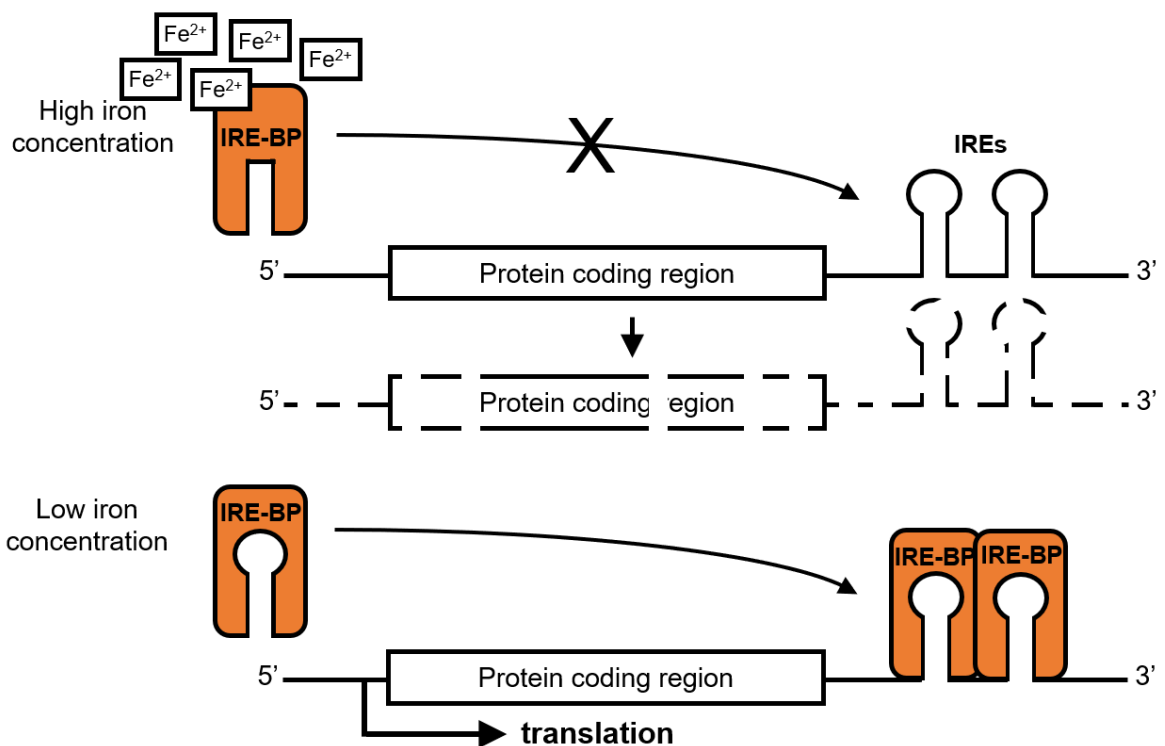


Fig. 6.1

- (a) (i) State the level of regulation shown in Fig. 6.1.

[1]

Translational level of regulation;;

- (ii) Explain how a change in cytosolic iron concentration results in an increased uptake of extracellular iron. [4]

1. When there is a decrease / low cytosolic iron concentration;
2. IRE-BP (is not bound by iron and therefore) is active;
3. Recognise and bind to IRE;
4. At 3' untranslated region;
5. Increases transferrin receptor mRNA stability / half-life;
6. mRNA is translated more frequently;
7. More transferrin receptor polypeptides synthesised
8. TfR proteins are brought to cell surface membrane;
9. via secretory vesicles;

Max 4

Ferritin proteins are involved in the storage of iron taken into the cell by TfRs.

The regulation of ferritin expression is shown in Fig. 6.2.

Ferritin mRNA in **cytosol**

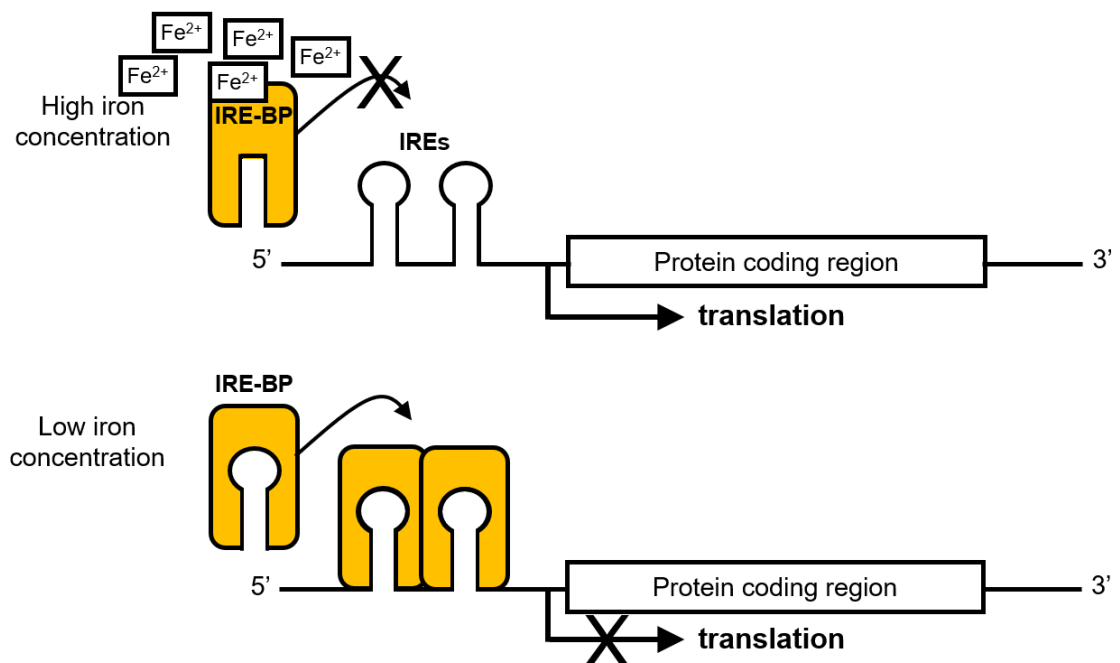


Fig. 6.2

- (b) Outline two differences in the regulation of ferritin and TfR expression. [2]

Comparison Feature	Regulation of expression of	
	Ferritin	Transferrin receptor (TfR)
1. effect of cytosolic iron conc. on expression	High [iron] results in increased expression of ferritin / translation of ferritin mRNA	High [iron] results in decreased expression of TfR / translation of TfR mRNA;; A: reverse argument
2. Effect of IRE-BP binding to IRE	Results in translational repression	Results in translational activation;; A: reverse argument
3. Effect on mRNA when no translation occurs	mRNA remains intact	mRNA is hydrolysed / broken down;;
4. Location where IRE-BP binds	In 5' UTR	In 3' UTR;;
5. AVP;;		

Accept any two

At a different level of regulation, activators and repressors act on distal elements to control the rate of transcription.

(c) Describe how activators alter the expression of a gene. [3]

1. Activators bind to enhancers;
2. Recruit DNA bending proteins;
3. Resulting in bending of DNA;
4. Activators also facilitates transcription initiation complex (TIC) assembly / RNA polymerase & transcription factors binding;
5. And facilitates correct positioning of TIC at promoter;
6. Leading to increased transcription rate;

[Total: 10]

- 7 Dachshunds have three basic coat types: wire-, smooth- or long-haired. These are affected by two genes, **H/h** and **G/g**. The presence of **H** always results in wire hair. When long-haired dogs are crossed amongst themselves, they always produce long-haired puppies.

Two groups of dogs heterozygous for both genes are crossed and the results from this cross are shown in Table 7.1.

Table 7.1

phenotype	number of offspring
wire-haired	90
smooth-haired	22
long-haired	8

Fig. 7.1 shows the pedigree for inheritance of coat type in Dachshunds.

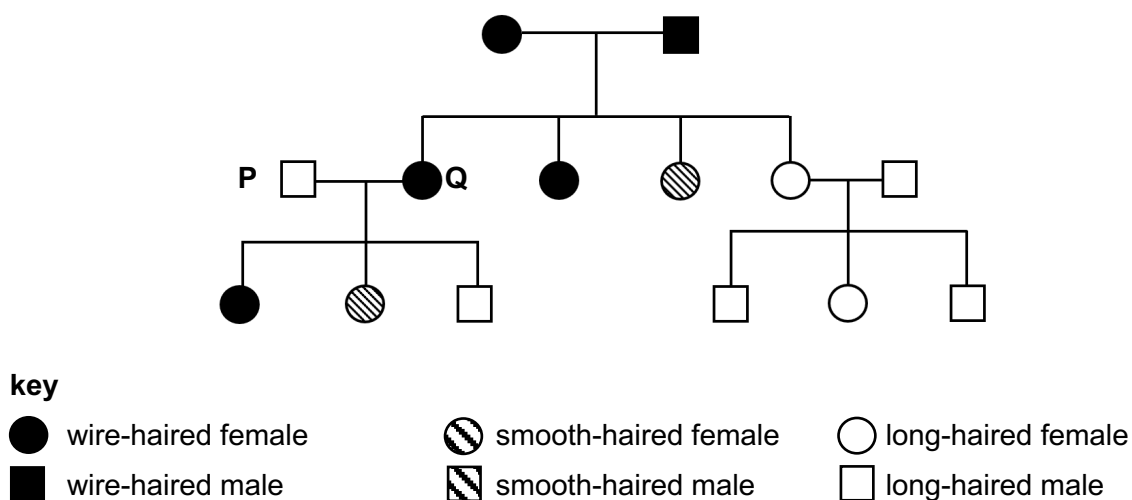


Fig. 7.1

- (a) Draw a genetic diagram in the space below showing the cross between **P** and **Q**.

Use the symbols given above and show all possible genotypes and phenotypes for the offspring of these parents.

[4]

Parental phenotypes	wire-haired female dog	x	long-haired male dog	
Parental genotypes	HhGg	x	hhgg	::
Gametes	HG Hg hG hg	x	hg	::

Punnett square	HG	Hg	hG	hg
hg	HhGg	Hhgg	hhGg	hhgg

	wire-haired	wire-haired	smooth-haired	long-haired
Offspring phenotypic ratio	2:		1:	1

offspring genotypes;;

offspring phenotypes;;

minus $\frac{1}{2}$ for each mistakes

- (b) In any genetic cross, the observed results are usually different from the expected results.

Suggest **one** reason why such a difference may occur, referring only to events after meiosis.

[1]

1. random fusion of gametes;;

2. differential survival of sperm/ ova (with particular genotypes);;

3. differential survival rate of zygotes (with particular genotypes);;

any one

Other than variation in coat type, dogs also display variation in their coat colours.

Table 7.2 shows the colour scale and number of dogs showing each of the phenotypes.

Table 7.2

colour						
colour scale	1	2	3	4	5	6
number of dogs	114	216	305	350	288	133

- (c) Distinguish between the two types of variation shown in coat type and coat colour in dogs.

[3]

	Discontinuous variation (coat type)	Continuous variation (coat colour)
1	discrete phenotypic classes ;;	a range of phenotypes R: no discrete class
2	Phenotype is controlled by one or two major genes;;	Phenotype is controlled by many genes (>2)

3	The environment has a small effect on the phenotype;;	The environment has a large effect on the phenotype
4	Effect of individual genes can be observed;;	Effect of individual polygenes cannot be observed

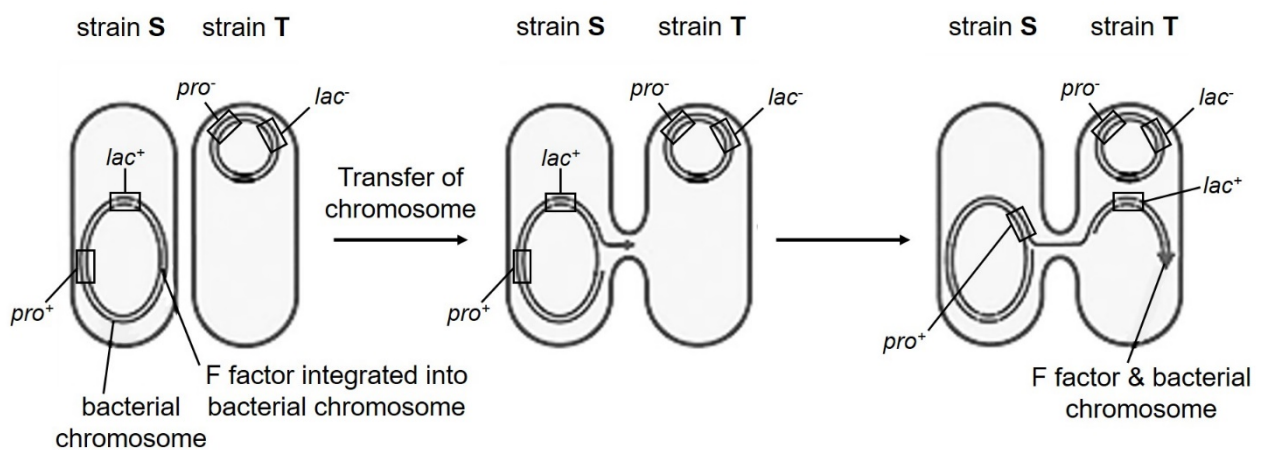
[Total: 8]

- 8 (a) (i) Using the *trp* operon as an example, explain what is meant by negative gene regulation. [1]
1. active form of the *trp* repressor;
 2. turns *trp* operon off;
- (ii) Explain why a single base-pair insertion in the regulatory gene of the *trp* operon may lead to an over-production of tryptophan. [5]
1. leads to frameshift / alteration of reading frame;
 2. change in mRNA codon sequence from point of insertion;
 3. change in amino acid sequence;
 4. polypeptide folds differently;
 5. change in 3-dimensional conformation of *trp* repressor;
- OR
1. leads to frameshift / alteration of reading frame;
 2. resulting in a stop codon;
 3. leads to premature termination of translation;
 4. producing truncated polypeptide;
 5. results in non-functional *trp* repressor / no *trp* repressor produced;
- OR
1. (insertion at promoter leads to) change in promoter sequence;
 2. change in shape of promoter sequence;
 3. shape of promoter sequence no longer complementary to DNA binding site of RNA polymerase / RNA polymerase cannot bind to promoter sequence;
 4. no transcription initiation complex assembled;
 5. no transcription of regulatory gene;

6. tryptophan unable to bind to allosteric site of *trp* repressor;
7. *trp* repressor always in the inactive form;
8. *trp* operon always switched on;
- A: *trp* repressor cannot bind to the operator
9. RNA polymerase binds to promoter continually;
10. for transcription and translation of structural genes to produce more enzymes involved in tryptophan synthesis;

An investigation was conducted to study gene transfer between two strains of *E. coli*, **S** and **T**. Bacterial cells of strain **S** are able to synthesise proline and metabolise lactose, while bacterial cells of strain **T** are unable to carry out both processes.

Fig. 8.1 shows the process of gene transfer during the investigation.



key

- pro⁺ : gene for proline synthesis present
 lac⁺ : gene for lactose metabolism present
 pro⁻ : gene for proline synthesis absent
 lac⁻ : gene for lactose metabolism absent

Fig. 8.1

- (b) (i) With reference to Fig. 8.1, describe the gene transfer process between *E. coli* strains **S** and **T**.

[3]

1. Bacterial conjugation;
2. strain **S** (F⁺ cell) synthesises a sex pilus;
3. and makes contact with recipient F⁻ cell from strain **T**;
4. forming a temporary mating bridge between the two cells;
5. transfer of bacterial chromosome containing F factor;

6. including pro⁺ and lac⁺ genes to recipient cell;

The two strains were mixed together in suspension in two separate experiments, for an incubation period of 10 minutes and 60 minutes respectively. At the end of the incubation, bacterial cells were then plated on an agar medium containing lactose and deficient in proline.

Table 8.1 shows the results of the two experiments.

Table 8.1

<i>E. coli</i> strain	colonies present in nutrient agar	
	after 10 minutes incubation	after 60 minutes incubation
S	94	95
T	42	93

- (ii) Explain why there is a larger number of *E. coli* strain **T** colonies when the two strains were incubated together for 60 minutes as compared to 10 minutes.

[2]

1. lac⁺ gene is transferred followed by pro⁺ gene;
2. Longer time for conjugation process;
3. recipient cell can acquire the pro⁺ gene;
4. to synthesise proline for protein synthesis;

[Total: 11]

- 9 Fig. 9.1 shows the phylogenetic relationship of four different species of snapping shrimps obtained based on comparison of nucleotide sequence.

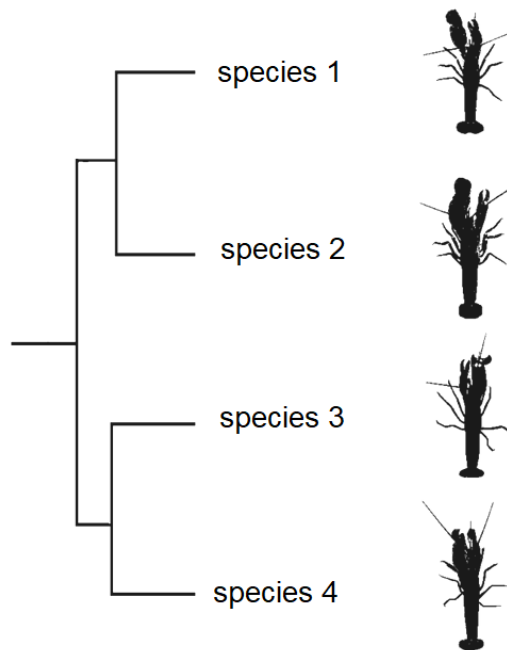


Fig. 9.1

- (a) Describe how molecular data was used to construct the phylogenetic tree. [2]

1. Comparison of DNA sequences of the same gene / homologous region;
2. from the different species of snapping shrimps;
3. High homology indicates that they are more closely related / share a more recent common ancestor (vice versa);
4. Indicated by a closer distance on phylogenetic tree;

Twenty million years ago, there was a gap between the continents of North America and South America through which the waters of the Caribbean Sea and Pacific Ocean flowed freely.

The snapping shrimps lived in this area between North America and South America.

About 3 million years ago, volcanic activity and sedimentation formed a narrow strip of land, Isthmus of Panama, joining North America and South America.

Fig. 9.2 shows the current distribution these four species.

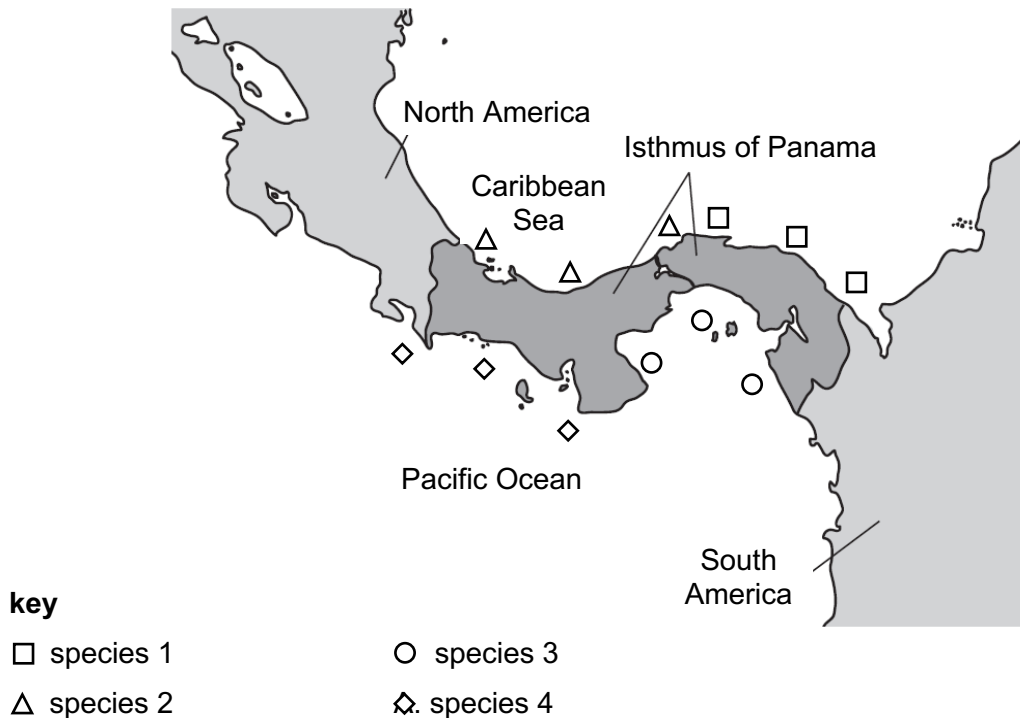


Fig. 9.2

Using information from Fig. 9.1 and Fig. 9.2,

- (b) (i) explain how biogeography and anatomical homology of snapping shrimps support Darwin's theory of evolution. [4]

1. (Fig. 9.2) more closely related species are found closer together;
2. compared with other species living in different areas;
3. evidence: species 1 and 2 are more closely related than with species 3 or 4;
4. (Fig. 9.1) similar phenotype shows common ancestry;
5. variations in phenotypes of snapping shrimps;
6. different selection pressures favouring different phenotypes;
7. evidence: variation in the claws;
8. (Fig. 9.2) allopatric speciation;
9. Due to no interbreeding between species 1 and 2 with species 3 and 4, and no gene flow;
10. evidence: separation by Isthmus of Panama

- (ii) suggest why breeding between species 1 and species 2 is possible. [1]

1. The habitat of species 1 and 2 overlaps / proximity (Fig 9.2);;
2. Species 1 and 2 are phenotypically similar (Fig 9.1);;

Any one

- (c) Explain how fossil records can also be used as evidence for evolutionary change of species. [2]

1. fossils show progressive changes in structure of organisms over time
2. showing intermediate forms;
3. fossils can be dated in chronological order;
4. some species disappeared showing extinction (as selected against);
5. new species appeared at a later stratigraphic level indicating speciation occurs;

[Total: 9]

- 10 (a) Describe the role of T lymphocytes in the immune response against *Mycobacterium tuberculosis* infection. [3]

1. Receptors on naïve helper T lymphocytes; in the lymph node
2. recognise antigens displayed by antigen presenting cell;
3. and are activated;
4. undergo clonal expansion;
5. migrate to the site of infection;
6. proliferate and surround tubercle bacilli;
7. forming a granuloma;

- (b) Explain **two** ways in which immunoglobulin differ from penicillin in response to infections by pathogen. [2]

Feature	Immunoglobulin	Penicillin
D1: Origin;;	Produced naturally by body's immune system	Produced by <i>Penicillium</i> mould
D2: Target;;	Effective against bacteria and viruses	Effective only against bacteria
D2: Function;;	Binds to specific pathogens to neutralise them / bring about opsonisation	Interferes with interpeptide linking of peptidoglycan, preventing cell wall synthesis

[Total: 5]

11 El Niño phenomenon is characterised by increased temperatures in the oceans.

(a) Explain how El Niño affects the corals. [2]

1. El Niño events trigger coral bleaching;
2. Rise in temperature causes the coral to expel the zooxanthellae / cause death of zooxanthellae;
3. Without nutrients and oxygen provided by zooxanthellae;
4. causing death of corals;

(b) (i) Describe how **one** human activity increases global sea temperatures. [1]

1. burning of fossil fuels/ deforestation /increase consumption of meat;
2. increased CO₂ /greenhouse gas emissions;

(ii) Suggest how reduction in the diversity of coral reefs reduces biodiversity. [2]

1. loss of habitat / refuge / spawning & nursing ground, for other marine species;;
or
loss of food source, affecting food chain;;
2. predation/death, leading to extinction of species;;

[Total: 5]